

15 a) Sketch the instrumentation of DTA and explain the Principle
Explain the DTA of $\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$ (8)

b) Briefly explain the principle and instrumentation of SEM
with the help of a diagram. (6)

ans: a) In DTA (Differential Thermal analysis), material and inert reference material (alumina, SiC or glass beads) are heated under identical conditions at a constant rate. The difference in temperature ΔT between the sample and reference is then plotted against temperature of the sample. It gives DTA curve or differential thermogram. Heat changes in the sample (exothermic or endothermic) can be detected.

Exothermic changes are represented by upward peak and endothermic changes are represented by downward peak.

DTA gives information on the transformations occur without the change of mass but involving heat change.
eg: Crystal Transition, glass transition and melting

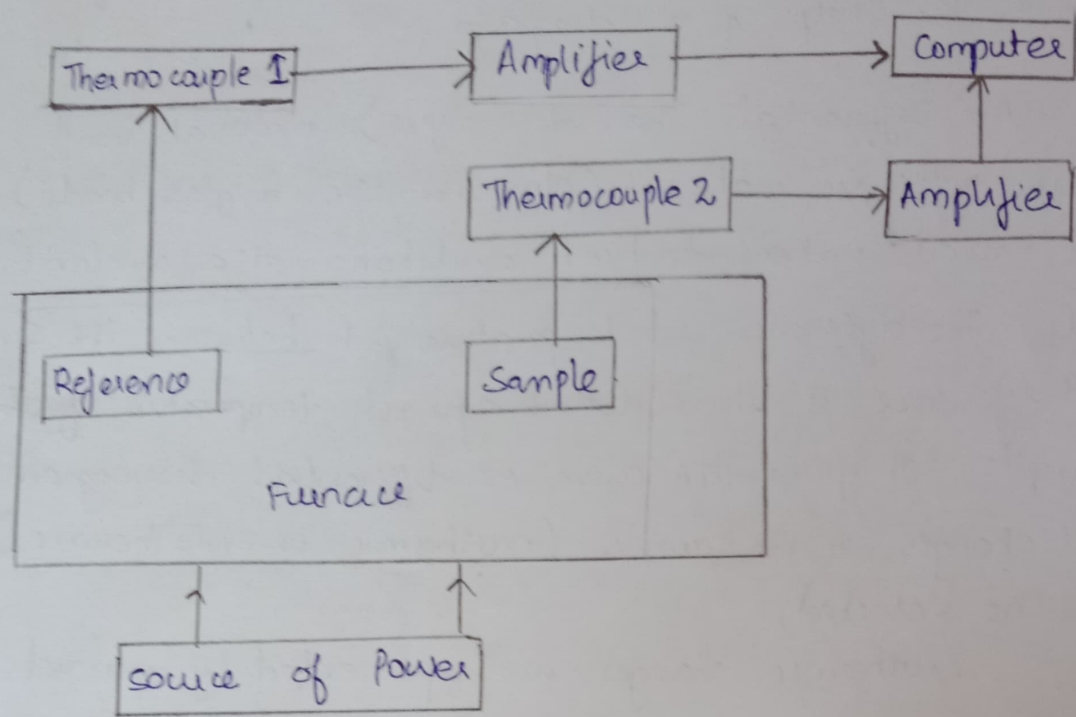
Area under a DTA peak is related to enthalpy change of the process

Physical changes such as transition, fusion, Vapourisation, sublimation etc are endothermic. Chemical changes like decomposition, dehydration are also endothermic

Oxidation in air, polymerisation etc are exothermic reactions

Peak area in differential thermogram is proportional to mass of the sample and enthalpy change (ΔH)

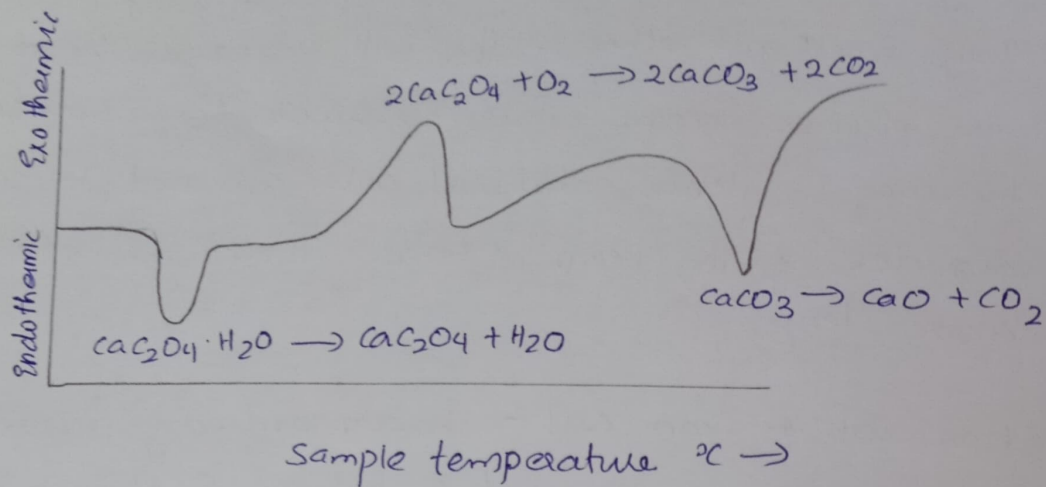
Instrumentation



Block diagram of DTA apparatus

Procedure

Sample and inert reference are heated under identical conditions at a constant rate. 2 Thermocouples T_1 and T_2 record the temperature in sample and reference. The data is amplified and processed in a computer.



b) SEM (Scanning electron Microscopy) is a surface characterization technique used in Nanotechnology. A Scanning electron Microscope is a powerful magnification tool that utilizes focussed beam of electrons to obtain information

Principle - SEM scans a focused electron beam over a surface to create an image. The electrons in the beam interact with the sample producing various signals that can be used to obtain information about surface topography and composition.

Instrumentation

1) Electron gun - Located at the top of the column where free electrons are generated by thermionic emissions from tungsten filament at 2700K . electrons are primarily accelerated towards an anode that is adjustable from 200V to 30kV

2, Condensing lens - All the electrons are negatively charged so they repel each other. This will increase the diameter of the electron beam which reduces the resolution of the image. Hence electrical coils are used to squeeze the diameter of the beam $< 5\text{nm}$. These are called condensing lens.

3, Deflection coil or Scan coil - In an ordinary microscope, the entire sample is illuminated but in SEM only a tiny spot is illuminated by the tiny beam. So the beam is moved in XY direction in small steps by a process called rastering. This is achieved by applying electric field. Entire sample is analysed by scanning the electron beam. Hence the name Scanning electron Microscope.

4, Objective lens - After the deflection coil another electromagnetic lens called objective lens is used. It focuses the electron beam down to the sample.

5, Sample chamber - After passing through the objective lens, electron beam passes into the sample chamber. This chamber holds the sample under vacuum to eliminate the interference of unwanted particles.

6, Detector - It is used to produce magnified images and collect other data. They will detect various signals given off by the sample. These signals include

secondary electrons (SE), back Scattered electrons (BSE) and x-rays.

working : when the electrons interact with sample, energetic electrons are released from sample surface.

The signals produced from the surface are

1, SE - Originate from the surface of sample. It give information on topography and composition.

2, X-rays - emitted from beneath the sample surface. Give information element and mineral.

3, BSE - Incident electrons reflected backwards.

It comes from deeper regions of sample. It give information on element and compound.

Give information on topography.

Answer the following

- 16 Describe the principle and steps to be followed in column chromatography.
- b. How are nanomaterials classified on the basis of dimension.

Ans. Principle : Components of a mixture are separated depending on the extent of adsorption on the stationary phase and the rate at which they are carried by the mobile phase through the stationary phase.

Stationary phase - Solid (Silica, Alumina, CaCO_3 , $\text{Ca}_3(\text{PO}_4)_2$)

Mobile phase - Liquid (Solvent like hexane)

For good separation polarity or activity of adsorbent should be high and polarity of solvent should be low.

Procedure;

- Adsorbent is packed in a cylindrical tube and plugged at the bottom by a piece of glass wool.

The mixture to be separated is dissolved in a suitable solvent and introduced from the top.

- As the mixture moves down, components are adsorbed at different regions depending on their adsorption affinity.
- Components with greater adsorption will be adsorbed at the top and others at the bottom.
- Different components are desorbed and collected separately by adding more solvent from the top and this process is known as elution. and the solvent used is known as eluent.
- Different components are collected separately and the solvent evaporated to get the pure components.

b. Based on the number of dimensions which are outside nanoscale; they are classified as,

- a) Zero dimension - All the 3 dimensions are in nanometric range. eg:- nanoparticles.
- b) One dimension - One of the dimension is outside

nanometric range. eg:- Nanowires, nanotubes, nanofibers

c) Two dimension - 2 of the dimensions are outside nanometric range and one is within the range.

eg: Nanofilms, nanolayer.

d) Three dimensions - Here all the dimensions are outside the nanometric range.

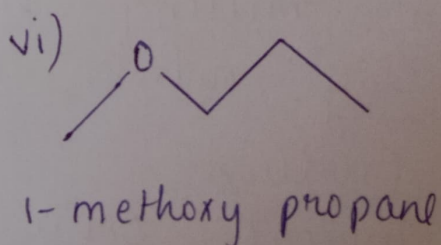
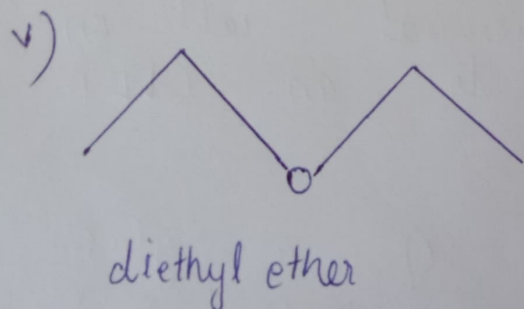
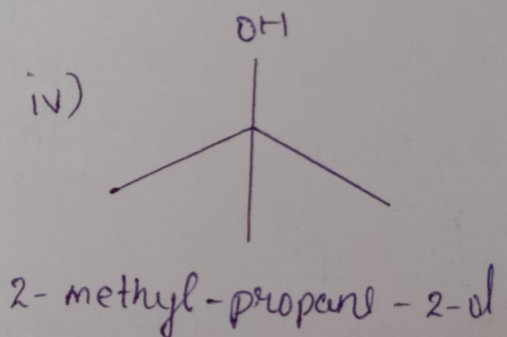
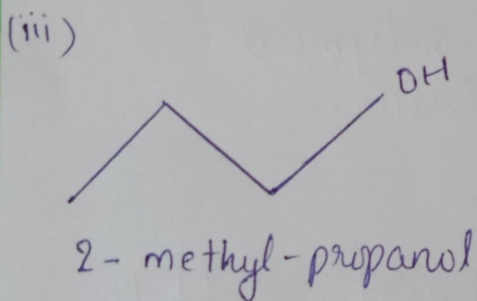
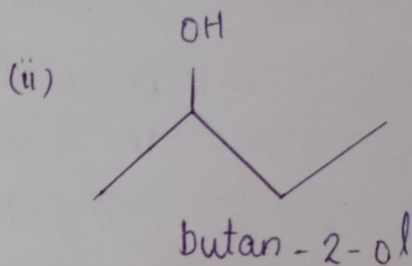
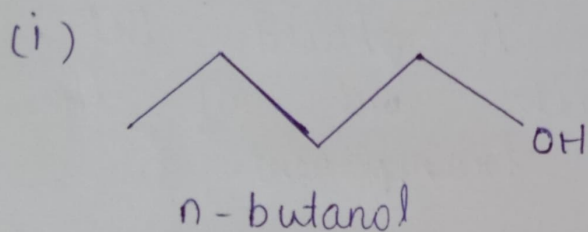
eg:- Bundles of nanowires and tubes, multilayers.

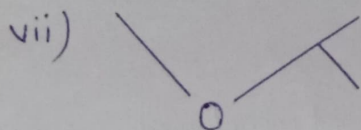
MODULE - IV

17 a) How many isomers (both structural and stereoisomers) are possible for $C_4H_{10}O$? Draw the structure of each.

b) What are OLEDs? Explain the construction and working of OLEDs.

Ans:- a) 8 isomers including optically active enantiomer pair,





2-methoxy propane

b) OLED (Organic Light Emitting Diode)

It is an LED in which the emissive electroluminescent layer is a film of organic compound which emit light in response to electric current. This organic semiconductor is placed bet 2 electrodes in which one of the electrode is made transparent.

There are 2 types of OLED

- ① which uses conducting polymers
- ② which use small molecule.

Principle

In conjugated polymer based OLED principle is HOMO-LUMO transitions.

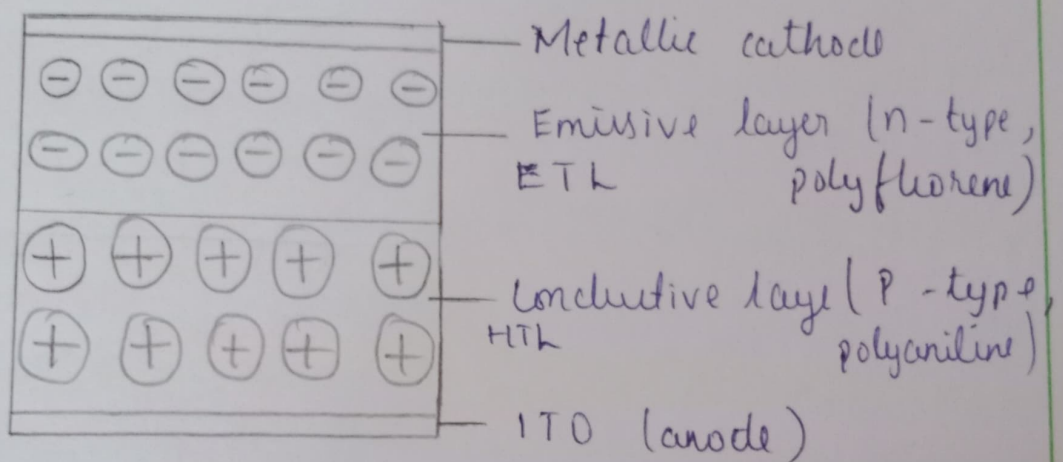
In small molecule OLED - Creation of light emitting electrochemical cell on adding mobile ions to an OLED

Construction

A typical OLED consist of a glass substrate, Indium Titanium Oxide (ITO) anode and a metallic cathode.

An organic bilayer is diffused into metallic cathode. An organic bilayer consist of a hole transport layer (HTL) on conductive layer and Electron Transport layer (ETL) or Emissive layer. Emissive layer is made of n-type polymer, commonly used is polyfluorene and conductive layer is made of p-type polymer.

Working



1. Apply a potential difference across anode and cathode.
2. As electricity starts to flow, cathode receives electrons from power sources and anode loses it.
3. Then the added electrons make emissive layer negatively charged, while the p-type conductive layer becomes positively charged.
4. Positive holes are much more mobile

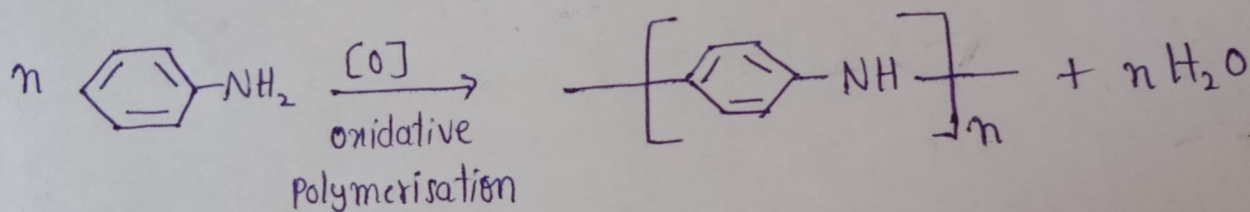
than negative electrons. So they jump across the boundary from conductive layer to the emissive layer.

5. When a hole meets an electron, the 2 things cancel out and release a brief burst of energy in the form of photons. This process is called recombination. It happens many times within a second. Thus OLED produces continuous light as long as current keeps flowing.

Q.19 How is polyaniline synthesized? List any two properties & applications.

(b)
Ans: • Chemical Synthesis: Aqueous solution of ammonium persulphate is added slowly to a solution of aniline dissolved in dil. HCl at a temperature of 5°C . After an hour a ppt. is formed, which can be removed by filtration. Polymer obtained is washed with NH_4OH solution & dried.

• Electrochemical Synthesis: Thin film of polymer is synthesized by electrochemical oxidation. It is done by dipping Pt foil electrodes in acidified aqueous solution of aniline at a potential of 0.7 V.



• Properties.

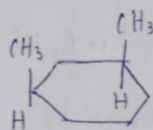
- used in sensors and electrochromatic devices due to its colour change in different oxidation states.
- Polyaniline is more noble than copper & less noble than silver.

• Applications.

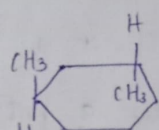
- used as electrode material in rechargeable batteries.
- Used in biosensors and solar energy absorption.

- (a) Draw the cis trans isomers of 1,3-dimethyl cyclohexane, which will be optically active. Draw all conformers, which conformer is more stable and why?

Ans:



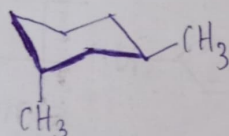
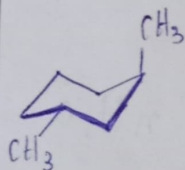
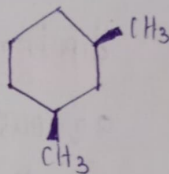
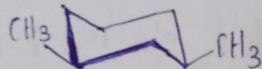
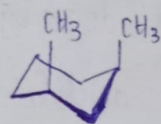
cis 1,3-dimethyl
cyclohexane



Trans 1,3-dimethyl
cyclohexane.

Trans 1,3-dimethyl cyclohexane is optically active since it does not possess plane of symmetry.

Conformers of 1,3-dimethyl cyclohexane.

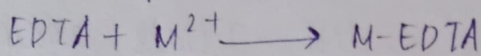


out of this, cis, 1,3-dimethyl cyclohexane in which two methyl group at equatorial position is most stable due to minimum interaction between the two groups.

Module - V

19 a) Explain the EDTA method for the estimation of hardness of water with calculation steps. (10)

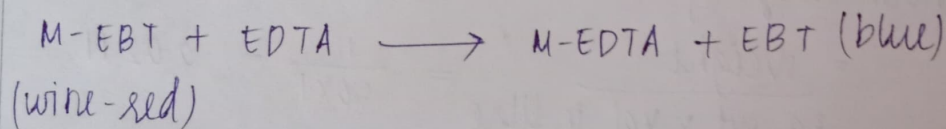
- EDTA complexes with metal ions M^{2+} present in water



Titration is conducted using EBT indicator

Principle

EBT forms a wine red coloured complex with M^{2+} at pH around 10 (M-EBT). As hard water is titrated, it is 1st complexed with M^{2+} ions in water. Towards the end point, no free metal ions, so it takes M^{2+} ions from M-EBT. M-EBT is less stable than M-EDTA. So free indicator (blue colour) generated at the end point. So colour of solution changes from wine red to blue.



$\text{NH}_4\text{Cl} - \text{NH}_4\text{OH}$ buffer solution is used to maintain pH around 10, since only at this pH EBT can form complex with Mg^{2+} & Ca^{2+}

Procedure

① Preparation of standard hardwater - About 1g of CaCO_3 is dissolved in minimum quantity dil HCl & evaporated to dryness in a china dish & the residue is dissolved in distilled water & made up to 1 litre.

- ② Standardisation of EDTA - 20 ml standard hardwater 2ml buffer & 2-3 drops of EBT are taken in a conical flask. Titrate against EDTA. let the volume be V_1 ml.
- ③ Estimation of total hardness - 20 ml unknown hardwater is pipetted out into a conical flask, 2ml buffer & 2-3 drops of EBT are added. Titrate against EDTA, let the volume be V_2 ml.
- ④ Estimation of permanent hardness - 250 ml unknown hardwater is boiled till volume reduces to 50 ml. So all bicarbonates will decompose to carbonate & hydroxides of Ca^{2+} & Mg^{2+} . Then the water is filtered and filtrate is made upto 250 ml with distilled water. 20 ml of this water is titrated against EDTA, let the volume be V_3 ml.

Calculation

- ① Normality of standard hardwater

$$= \frac{\text{Weight}}{\text{Eq. wt} \times \text{Vol. in litre}} = \frac{1}{50 \times 1} = 0.02 \text{ N}$$

- ② Standardisation of EDTA

20 ml std. hard water $\equiv V_1$ ml N_2 normal EDTA

($V_1 N_1 = V_2 N_2$ general equation)

$$\therefore \text{Normality of EDTA, } N_2 = \frac{20 \times 0.02}{V_1}$$

- ③ Estimation of total hardness

20 ml unknown hardwater $\equiv V_2$ ml N_2 normal EDTA

$$\therefore \text{Normality of hardwater, } N_3 = \frac{V_2 \times 20 \times 0.02}{20 \times V_1}$$

$$= \frac{V_2 \times 0.02}{V_1}$$

total hardness = Normality $\times$ Eq. Wt of CaCO_3 g/L

$$= \frac{V_2 \times 0.02}{V_1} \times 50 \text{ g/L}$$

$$= \frac{V_2 \times 0.02 \times 50 \times 1000}{V_1} \text{ mg/L}$$

$$= \frac{V_2}{V_1} \times 1000 \text{ mg/L}$$

Estimation of permanent hardness

20 ml, N_4 normal boiled hardwater $\equiv$ V_3 ml N_2 normal EDTA

$$20 \times N_4 = \frac{V_3 \times 20 \times 0.02}{V_1}$$

$$\therefore N_4 = \frac{V_3 \times 0.02}{V_1}$$

$$\therefore \text{permanent hardness} = \frac{V_3}{V_1} \times 1000 \text{ mg/L}$$

$$\left(\text{ie, } \frac{V_3}{V_1} \times 50 \times 0.02 \times 1000 \text{ mg/L} \right)$$

$$\therefore \text{Temporary hardness} = \left(\frac{V_2 - V_3}{V_1} \right) \times 1000 \text{ mg/L or ppm}$$

$$\left\{ \begin{array}{l} \text{Temporary} \\ \text{hardness} \end{array} \right. = \text{Total} - \text{Permanent}$$

write the procedure for estimating COD of a sample waste water. (4)

COD is a measure of all oxidisable impurities, biologically active and biologically inert, present in 1L of sewage water. $\therefore \text{COD} > \text{BOD}$

It is the amount of O_2 required for the complete oxidation of biologically active and biologically inert materials present in 1L of sewage water using chemical oxidizing agents such as $\text{K}_2\text{Cr}_2\text{O}_7$

It needs 3688.

procedure

100 ml sewage water is taken in an RB flask.
Mixed with 20 ml, 0.1 N $K_2Cr_2O_7$ & 20 ml 1N H_2SO_4 .
Reflux for 2 hrs in presence of small amount of
 Ag_2SO_4 catalyst. Solution is cooled & titrated with
std. Ferrous ammonium sulphate in a burette.
A blank is conducted with pure water

$$\text{Then COD} = \left(\frac{V_b - V_t}{V_e} \right) \times N \times 8 \times 1000 \text{ ppm}$$

$V_b \rightarrow$ vol. FAS in blank

$V_t \rightarrow$ vol. of FAS in test solution

$V_e \rightarrow$ vol. of effluent taken

(FAS - Ferrous
Ammonium
sulphate)

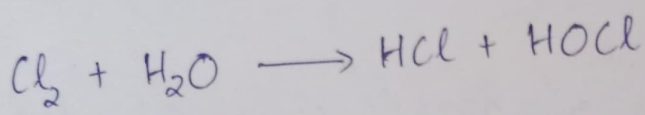
CHEMISTRY

20. (a) Discuss the action of chlorine as a disinfectant. How is it applied? What is break point chlorination? Write any two advantages of break point chlorination. (10)

(b) A pure water sample is added with 90g carbohydrate (CH_2O) per litre find the maximum BOD possible for the water sample. (4)

ANSWER:

(a) Chlorine when added to water in gaseous form or concentrated solution form, it produces HOCl .



HOCl kills the microorganisms by deactivating the enzymes.

Chlorinator is the apparatus used for chlorination, which utilises liquid chlorine. It is a high tower with a no. of baffle plates. Water along with required quantity of concentrated chlorine solution are introduced from the top. As they move down, thorough mixing occurs and disinfection is effected. Outlet is provided at the bottom.

Break point chlorination:

Addition of Cl_2 to water in amounts sufficient to kill all microorganisms & destroy completely by oxidation.

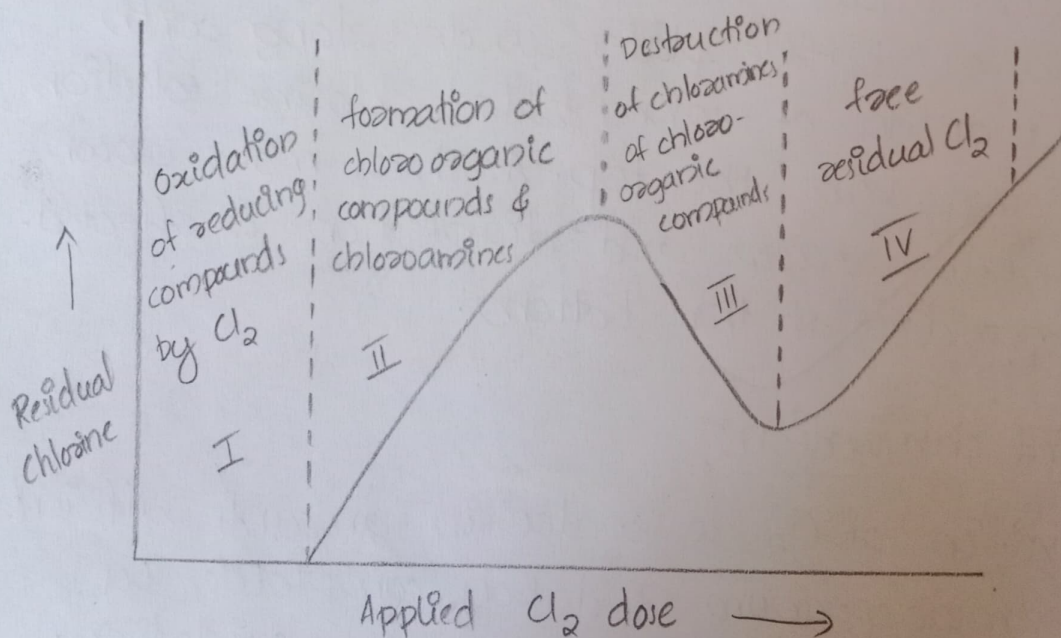
Impurities like organic matter, reducing substances and colouring matter, free ammonia get oxidised and destroyed. Water will contain 0.1 - 0.2 ppm free Cl_2 to continue disinfection during storage & transport.

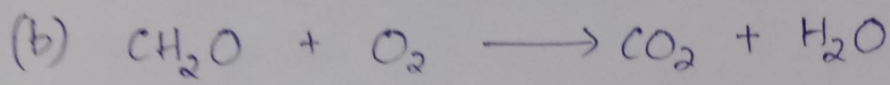
Addition of Cl_2 at the dip or break is called break point chlorination. Here residual chlorine begins to appear.

All taste & odour disappear at break point. Residual Cl_2 is a powerful disinfectant which can destroy all types of pathogenic organisms present in water.

Advantages:

- (i) Completely oxidises all organic compounds, NH_3 and other reducing compounds.
- (ii) Destroys all the disease producing bacteria.





30 mg carbohydrate react with 32 mg Oxygen

$$\text{So; } 1 \text{ mg} \longrightarrow \frac{32}{30} \text{ mg}$$

$$\text{Here, } 90 \text{ mg} \longrightarrow \frac{32}{30} \times 90$$

$$= 96 \text{ mg.}$$

Therefore max. BOD required = 96 mg